

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
In [2]: df = pd.read_csv("data/raw/monthly-housing.csv")
df.head(3)
```

```
Out[2]:
```

	date	hpi	numsold
0	1/1/1991	100.00	30
1	2/1/1991	100.48	40
2	3/1/1991	100.74	51

Remove null value

```
In [3]: print(df.shape)
print(df[["date", "hpi"]].isnull().sum())
```

```
(280, 3)
date    12
hpi     12
dtype: int64
```

```
In [4]: df_clean = df.dropna(subset=["date", "hpi"]).copy()
print(df_clean[["date", "hpi"]].isnull().sum())
print(df_clean.shape)
```

```
date     0
hpi      0
dtype: int64
(268, 3)
```

Change date format

```
In [5]: df_clean["date"] = pd.to_datetime(df_clean["date"])
df_clean = df_clean.sort_values("date").reset_index(drop=True)
df_clean = df_clean.set_index("date")
df_clean = df_clean.asfreq("MS") # set as the beginning of the month
```

```
In [6]: print(df_clean.index)
print(df_clean.head(3))
print(df_clean.tail(3))
```

```
DatetimeIndex(['1991-01-01', '1991-02-01', '1991-03-01', '1991-04-01',
               '1991-05-01', '1991-06-01', '1991-07-01', '1991-08-01',
               '1991-09-01', '1991-10-01',
               ...
               '2012-07-01', '2012-08-01', '2012-09-01', '2012-10-01',
               '2012-11-01', '2012-12-01', '2013-01-01', '2013-02-01',
               '2013-03-01', '2013-04-01'],
              dtype='datetime64[ns]', name='date', length=268, freq='MS')
   hpi  numsold
date
1991-01-01  100.00      30
1991-02-01  100.48      40
1991-03-01  100.74      51
              hpi  numsold
date
2013-02-01  194.31      36
2013-03-01  198.06      41
2013-04-01  200.87      43
```

Visualize df_clean

```
In [7]: plt.figure(figsize=(10, 4))

plt.plot(
    df_clean.index,
    df_clean["hpi"],
    linewidth=0.8
)

y_min = df_clean["hpi"].min()
y_max = df_clean["hpi"].max()
padding = (y_max - y_min) * 0.03

plt.ylim(y_min - padding, y_max + padding)

plt.title("Housing Price Index Over Time")
plt.xlabel("Date")
plt.ylabel("HPI")
plt.grid(True, alpha=0.3)

plt.show()
```



save df_clean

```
In [8]: df_clean.to_csv("data/processed/monthly_housing_clean.csv", index=True)
```

train test split

```
In [9]: train = df_clean.iloc[:-12].copy()
test = df_clean.iloc[-12:].copy()

print("Train shape:", train.shape)
print("Test shape:", test.shape)

print(train.head(3))
print(train.tail(3))

print(test.head(3))
print(test.tail(3))
```

Train shape: (256, 2)

Test shape: (12, 2)

date	hpi	numsold
1991-01-01	100.00	30
1991-02-01	100.48	40
1991-03-01	100.74	51
hpi numsold		
date		
2012-02-01	181.36	30
2012-03-01	184.15	34
2012-04-01	187.26	34
hpi numsold		
date		
2012-05-01	190.29	35
2012-06-01	192.25	34
2012-07-01	192.66	33
hpi numsold		
date		
2013-02-01	194.31	36
2013-03-01	198.06	41
2013-04-01	200.87	43

```
In [10]: train.to_csv("data/processed/monthly_housing_train.csv", index=True)
test.to_csv("data/processed/monthly_housing_test.csv", index=True)
```